Sagnik Datta Bhowmick

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EDUCATION

Institute of Engineering and Management (IEM), Kolkata 2025 – 2029
Bachelor of Technology in Computer Science and Engineering *CGPA: 9.31*

Indian Institute of Technology Madras (IIT Madras) 2026 – Present
Bachelor of Science in Data Science and Applications (Foundation Level) *CGPA: 10.00/10.00*

RESEARCH EXPERIENCE

Centre of Excellence for Data Science (CEDS), IEM Kolkata 2026 – Present
Student Member

- Contributing to faculty-mentored research in computer vision and deep learning, focusing on semantic perception systems for autonomous vehicle applications.
- Designing and executing experiments in PyTorch to benchmark semantic segmentation architectures, analyzing the performance tradeoffs between classical Fully Convolutional Networks and modern transformer models.

SELECTED TECHNICAL PROJECTS

Sequential Reciprocal Matching under Extreme Delay | *Python, Operations Research, Decision Theory*

- Formulated a resource-constrained sequential matching policy utilizing Partially Observable Markov Decision Processes (POMDP) and Value of Information (VoI) heuristics.
- Achieved a 51.9% improvement in Mutual Second-Meeting Intention (MSMI) from 0.258 to 0.392 over the greedy baseline across a rigorous 10-seed simulated evaluation covering six environmental scenarios.

Road Segmentation for Autonomous Driving | *PyTorch, Computer Vision, Hugging Face*

- Systematically evaluated deep learning architectures for binary semantic road segmentation, progressing from a from-scratch FCN-ResNet50 (87.11% validation IoU) to fine-tuned transformer backbones.
- Adapted an ADE20K-pretrained SegFormer-B2 model, applying layer-wise learning rates to backbone stages 2 and 3, achieving a validation IoU of 92.95% and a Dice Score of 96.34%.

CiteCheck: AI Citation Verification Framework | *FastAPI, NLP, Python, React*

- Engineered an applied AI backend utilizing Gemini to parse unstructured document bibliographies and deterministically verify claims against Crossref, OpenAlex, and Semantic Scholar APIs.
- Validated extraction accuracy and mitigated hallucination rates using a custom evaluation framework comprising 145 deterministic boundary-tested cases.

TinyURL & DSA Laboratory | *Cloudflare Workers, D1/SQLite, React, C11*

- Built an edge-native URL shortener featuring dual-tier KV caching, Base62 encoding, sliding-window rate limiting, alongside an interactive React Hash Table visualization lab backed by a memory-safe C11 reference implementation.

Ingredient Label Checker | *Python, Playwright, OpenRouter, NVIDIA Nemotron*

- Developed a browser extension for complex dietary profiling (Vegan, Jain, allergens) combining dynamic Playwright webpage extraction with deterministic rules and LLM-based fallback classification.

RentX: Peer-to-Peer Rental Marketplace | *Next.js, TypeScript, PostgreSQL, Prisma, Socket.IO*

- Architected a hyperlocal rental platform featuring a digital Item Passport, multidimensional reputation metrics, real-time messaging, and an auditable off-chain state machine for deposit lifecycles.

AI-Based Spinal Posture Detection | *OpenCV, Scikit-learn, MediaPipe*

- Built a real-time computer vision pipeline to extract frontal and lateral body landmarks, calculating critical geometric postural angles for a serialized Scikit-learn Random Forest classifier.

TECHNICAL SKILLS

- **Languages:** Python, C, C++, TypeScript, JavaScript, SQL
- **AI/ML & Vision:** PyTorch, Scikit-learn, OpenCV, MediaPipe, Hugging Face, LLM Integrations
- **Backend & Web:** FastAPI, Next.js, React, Node.js, Cloudflare Workers, PostgreSQL, SQLite, Socket.IO
- **Tools & Theory:** Linux, Git, Docker, Playwright, Optimization, Decision Theory (POMDP), Data Structures

ACHIEVEMENTS

- **Second Place – INNOV12 Buildathon:** Secured second place for developing the Ingredient Label Checker, an AI-powered application for ingredient and allergen analysis.
- **Top 20 Projects – Innovacion:** Selected among the top 20 projects for the independent development of the AI-Based Spinal Posture Detection project.
- **20th Rank – CSE Department:** Secured 20th position in the Computer Science and Engineering department during my first year.

CERTIFICATIONS

- **NPTEL: The Joy of Computing using Python (IIT Ropar)** – Elite + Gold, 96% (Top 5% Topper) *Jan – Apr 2026*
- **NPTEL: Data Structures and Algorithms Design (IIT Kanpur)** – Elite + Silver, 79% *Jul – Dec 2026*
- **NPTEL: Programming in Modern C++ (IIT Kharagpur)** – Elite, 64% *Jul – Dec 2026*